

AL-AMEED MEDICAL COLLEGE

CLINICAL HISTORY TAKING & CARDIOVASCULAR SYSTEM (CVS)

Basic Structured Guide for 3rd Stage Medical Students

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Target Audience: 3rd Stage (Stage 3) Medical Students

Source: Macleod's Clinical Examination (15th Edition)

Learning Objectives

- Understand why taking a medical history is the most important clinical diagnostic tool.
- Memorize the 9 standard components of a medical history as outlined in Macleod's.
- Learn what specific information to collect under each history component.
- Apply the SOCRATES framework to analyze medical symptoms systematically.
- Learn the basic symptoms and risk factors of the Cardiovascular System (CVS).

Why History Taking Matters

- Primary Diagnostic Tool: Up to 80% of medical diagnoses are made from the history alone.
- Guides the Physical Exam: History tells you which body systems need detailed examination.
- Directs Investigation: Helps choose the right blood tests, X-rays, or ECGs without wasting time.
- Saves Time & Resources: Prevents unnecessary, costly, or painful medical tests.
- Core Student Skill: History taking is tested in every clinical rotation and OSCE exam.

The 9 Components of Medical History

**1. Personal Data
(Demographics)**

**2. Chief Complaint (CC) &
Duration**

**3. History of Present
Illness (HPI)**

4. Past Medical History

5. Past Surgical History

6. Drug History & Allergies

7. Family History

**8. Social History &
Lifestyle**

**9. Review of Other
Systems**

Mind Map: Full History Structure

THE MEDICAL HISTORY

1. Personal Data

2. Chief Complaint

3. Present Illness (HPI)

4. Past Medical History

5. Past Surgical History

6. Drug History & Allergies

7. Family History

8. Social History

9. Review of Systems

1. Personal Data (Demographics)

- Full Name & Age: Helps confirm patient identity and age-related disease probability.
- Sex / Gender: Certain diseases are more common in males or females.
- Occupation: Identifies work-related exposures and physical demands.
- Marital Status & Address: Provides context on social support and living location.
- Date & Time of Admission: Crucial for establishing disease timeline.
- Source of History: Patient themselves, relative, or medical transfer notes.

2. Chief Complaint (CC) & Duration

- Definition: The main symptom that caused the patient to seek medical help.
- Patient's Own Words: Record exactly what the patient says (e.g., 'chest pain', not 'angina').
- Include Duration: Always specify how long the symptom has been present.
- Example 1: 'Shortness of breath for 3 days'.
- Example 2: 'Severe headache for 2 hours'.
- Keep It Short: Should be 1 or 2 symptoms maximum, not a long paragraph.

3. History of Present Illness (HPI)

- Chronological Story: Tell the full story of the main symptom from onset until today.
- Symptom Details: When did it start? Was it sudden or gradual? Is it constant or intermittent?
- Associated Symptoms: What other symptoms happen at the same time? (e.g., pain with nausea).
- Previous Episodes: Has the patient ever experienced this exact symptom before?
- Patient's Thoughts (ICE): Ask what they think is wrong (Ideas) and what worries them (Concerns).

Symptom Analysis: SOCRATES

SOCRATES Letter	Simple Explanation & Question
S - Site	Where exactly is the pain or symptom located?
O - Onset	Did it start suddenly or develop gradually over hours/days?
C - Character	What does it feel like? (Sharp, dull, crushing, burning)
R - Radiation	Does the pain spread anywhere else? (e.g., arm or jaw)
A - Associated Symptoms	Are there other symptoms like sweating, nausea, or vomiting?
T - Timing / Duration	How long does it last? Is it constant or coming in waves?
E - Exacerbating / Relieving	What makes it worse (exercise)? What makes it better (rest)?
S - Severity	How severe is it on a scale from 1 to 10?

4. Past Medical History

- Purpose: Record all major illnesses the patient has had in the past.
- High-Yield Illnesses to Ask About Directly:
 - • Hypertension (High Blood Pressure)
 - • Diabetes Mellitus (High Blood Sugar)
 - • Asthma or Chronic Lung Disease
 - • Previous Heart Attacks or Stroke
 - • Kidney Disease or Liver Disease
- Hospital Admissions: Ask if they have ever been admitted to a hospital before.

5. Past Surgical History

- Surgical Operations: Record any surgical procedures performed in the past.
- Details to Note: Name of operation, year performed, and hospital.
- Complications: Ask if there were any bleeding or wound infection problems.
- Anesthesia Problems: Ask if the patient or family ever had severe reactions to anesthesia.
- Blood Transfusions: Note if they ever received blood transfusions in past surgeries.

6. Drug History & Allergies

- Current Medications: Name of drug, dosage, and frequency (e.g., Aspirin 100mg daily).
- Over-the-Counter Drugs: Ask about pain relievers, vitamins, and herbal remedies.
- Medication Adherence: Ask gently if they regularly take their prescribed medicines.
- Drug Allergies: Always ask 'Are you allergic to any medications?'
- Distinguish Allergy vs Side Effect: True allergy causes rash/swelling; side effect causes stomach upset.
- **RED FLAG: Document true penicillin or severe drug allergies clearly in red!**

7. Family History

- First-Degree Relatives: Health status of parents, brothers, sisters, and children.
- Cause of Death: If deceased, record cause and age at death.
- Inherited Conditions: Ask if heart disease, high BP, diabetes, or stroke runs in the family.
- Premature Heart Disease: Heart attacks in young family members (<55 years in men, <60 in women).
- Familial Patterns: Helps identify genetic or shared environmental health risks.

8. Social History & Lifestyle

- Smoking History: Do they smoke? How many cigarettes a day and for how many years?
- Alcohol Intake: How many drinks/units per week?
- Occupation & Work Environment: What is their job? Any exposure to dust/chemicals?
- Living Situation: Who do they live with? Do they live in a house or apartment with stairs?
- Diet & Exercise: Basic eating habits and level of daily physical activity.

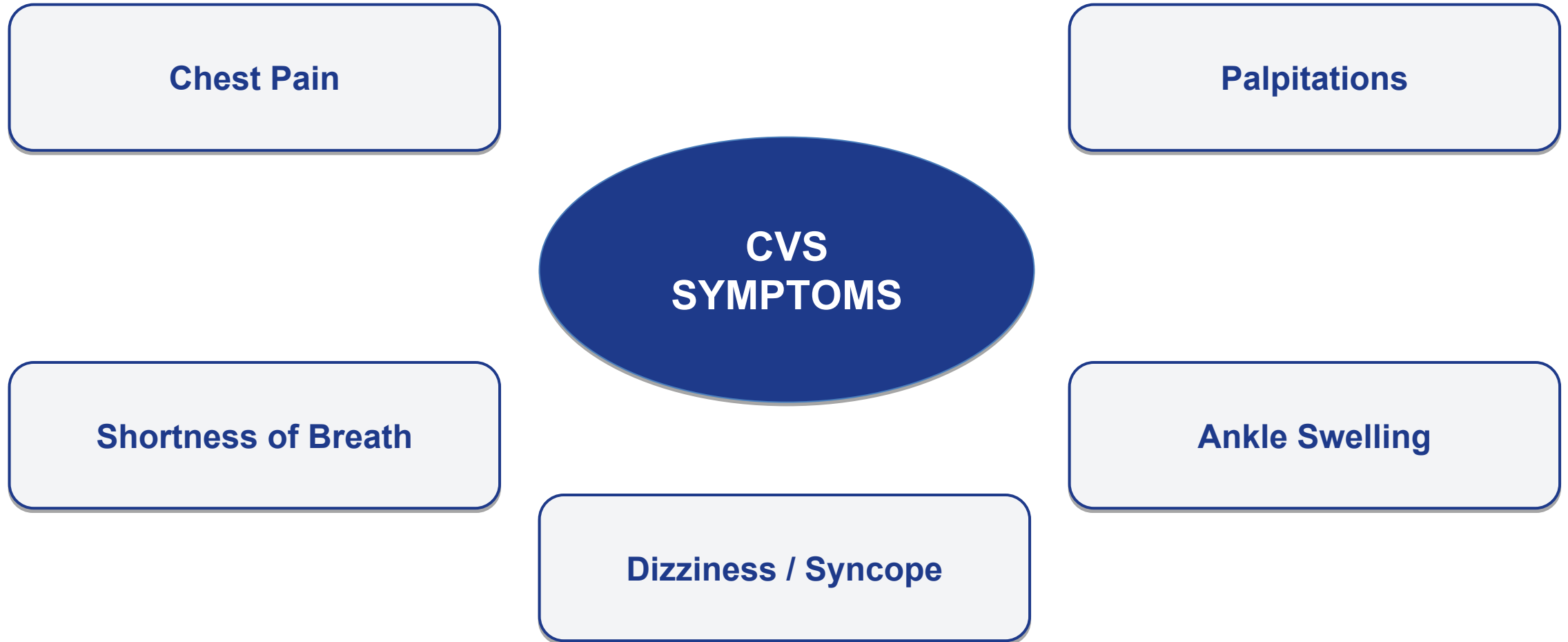
9. Review of Other Systems

- Purpose: A quick screening checklist to ensure no major symptom was missed.
- Cardiovascular: Chest pain, breathlessness, ankle swelling, palpitations.
- Respiratory: Cough, sputum, shortness of breath, wheeze.
- Gastrointestinal: Stomach pain, nausea, heartburn, bowel habit changes.
- Nervous System: Dizziness, faints, headache, weakness.
- Genitourinary & Musculoskeletal: Urination problems, joint or muscle pain.

SECTION 2: CARDIOVASCULAR SYSTEM (CVS) HISTORY

Brief Introductory Approach for 3rd Stage Students

Mind Map: Core CVS Symptoms



CVS Symptom 1: Chest Pain

- Common Cause: Lack of blood flow to the heart muscle (Ischemia).
- Angina (Stable): Dull heaviness or tightness across the chest, triggered by walking/exertion, relieved by rest.
- Heart Attack (MI): Severe crushing chest pain, lasts >20 minutes, accompanied by sweating and nausea.
- Where it Spreads: Pain often radiates to the left arm, neck, or jaw.
- **RED FLAG: Sudden excruciating 'tearing' chest pain spreading to the back is an emergency (Aortic Dissection)!**

Chest Pain: Cardiac vs Non-Cardiac

CARDIAC CAUSES

- Heavy, crushing, or squeezing feeling
- Located centrally behind breastbone
- Brought on by physical exercise or stress
- Eased by rest or nitrate spray
- May spread to jaw or left arm

NON-CARDIAC CAUSES

- Muscle / Bone: Sharp pain, worse when touching chest wall or moving
- Lung / Pleuritic: Sharp pain, worse when taking a deep breath or coughing
- Stomach / Oesophagus: Burning heartburn feeling after eating

CVS Symptom 2: Shortness of Breath

- Dyspnoea on Exertion: Feeling breathless when walking or climbing stairs.
- Orthopnoea: Shortness of breath when lying flat in bed.
- • Key Question: 'How many pillows do you sleep on to breathe comfortably?'
- Paroxysmal Nocturnal Dyspnoea (PND): Waking up suddenly at night gasping for air.
- Why it Happens: Fluid backs up into the lungs because the heart muscle is weak (Heart Failure).

CVS Symptom 3: Palpitations & Dizziness

- Palpitations: Unpleasant awareness of the heartbeat (e.g., 'fluttering', 'racing', or 'skipped beats').
- Key Question: Ask the patient to tap out the rhythm with their finger.
- Dizziness & Fainting (Syncope): Temporary loss of consciousness due to reduced blood flow to the brain.
- Postural Dizziness: Dizziness when standing up quickly (low blood pressure).
- **RED FLAG: Fainting during physical exercise requires urgent heart evaluation!**

CVS Symptom 4: Ankle Swelling & Leg Pain

- Ankle Swelling (Edema): Fluid building up in both legs/ankles due to heart failure.
- Pitting Edema: Pressing a thumb on the ankle leaves a visible dent.
- Intermittent Claudication: Cramp-like leg pain when walking a short distance that stops with rest.
- Cause of Claudication: Narrowed arteries reducing blood flow to leg muscles (Peripheral Arterial Disease).
- Simple Question: 'Do your legs hurt when walking, and does rest make the pain go away?'

Cardiovascular Risk Factors

- Smoking: Major cause of coronary artery and peripheral vascular disease.
- High Blood Pressure (Hypertension): Damages blood vessel walls over time.
- Diabetes Mellitus: High blood sugar damages heart blood vessels (can cause 'painless' heart attacks).
- High Cholesterol (Dyslipidaemia): Causes fatty buildup (plaques) in arteries.
- Family History: Early heart disease in close family members.
- Unhealthy Lifestyle: Lack of physical exercise, obesity, and high-fat diet.

Sample CVS History Script

Student: 'What brings you to the hospital today?'

Patient: 'I get this heavy pressure in my chest when I walk up stairs.'

Student: 'Does the pressure spread anywhere else?'

Patient: 'Yes, sometimes up into my jaw and left shoulder.'

Student: 'What happens when you stop and rest?'

Patient: 'It goes away completely after 3 or 4 minutes.'

Student: 'Do you ever feel short of breath or notice swelling in your ankles?'

Patient: 'I get a bit short of breath on hills, but no ankle swelling.'

Common Student Mistakes to Avoid

- Mistake 1: Writing medical diagnosis in Chief Complaint (e.g., write 'chest pain', not 'angina').
- Mistake 2: Forgetting to record duration of the main symptom.
- Mistake 3: Skipping Past Medical History questions like diabetes or high blood pressure.
- Mistake 4: Not asking about drug allergies.
- Mistake 5: Asking leading questions (e.g., ask 'What is the pain like?', NOT 'Is the pain sharp?').

SUMMARY

Summary & Key Take-Home Points

- History taking is the foundation of diagnosis — up to 80% of diagnoses come from history.
- Always follow the 9 standard components in order from Personal Data to Systems Review.
- Use SOCRATES to analyze any symptom thoroughly.
- CVS history focuses on 5 key symptoms: Chest Pain, Breathlessness, Palpitations, Swelling, and Syncope.
- Always screen for major CVS risk factors: Smoking, High BP, Diabetes, Lipids, and Family History.

QUESTIONS

Self-Assessment MCQs (1 - 3)

Q1. What percentage of medical diagnoses are made from history taking alone?

A) 20% | B) 50% | C) 80% | D) 95%

Answer: C (80%)

Q2. Which section of the history includes recording 'Chest pain for 2 hours'?

A) Personal Data | B) Chief Complaint | C) Social History | D) Systems Review

Answer: B (Chief Complaint)

Q3. What does the letter 'R' stand for in the SOCRATES mnemonic?

A) Rest | B) Radiation | C) Relieving factors | D) Risk

Answer: B (Radiation)

QUESTIONS

Self-Assessment MCQs (4 - 5)

Q4. Shortness of breath when lying flat in bed is called:

A) Exertional dyspnoea | B) Orthopnoea | C) Claudication | D) Palpitations

Answer: B (Orthopnoea)

Q5. Which of the following is a major risk factor for cardiovascular disease?

A) Low blood pressure | B) Cigarette smoking | C) Exercise | D) Young age

Answer: B (Cigarette smoking)

References & Recommended Reading

- Primary Source Text:
 - • Dover AR, Innes JA, Fairhurst K, eds. Macleod's Clinical Examination. 15th ed. Elsevier; 2024.
- Core Chapters:
 - • Chapter 1: Managing clinical encounters with patients.
 - • Chapter 2: General aspects of history taking.
 - • Chapter 4: The Cardiovascular System.
- Al-Ameed Medical College — Clinical Examination Curriculum for 3rd Stage Students.